Saturating Splines and Feature Selection

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The problem / what we want

History of previous work

Saturating splines - what they are

Reformulating the optimization - atomic measures, generalized additive models (GAMs)

Solving the optimization problem - conditional gradient descent, backfitting

Experiments - low error, variable selection, benefits of a general loss function

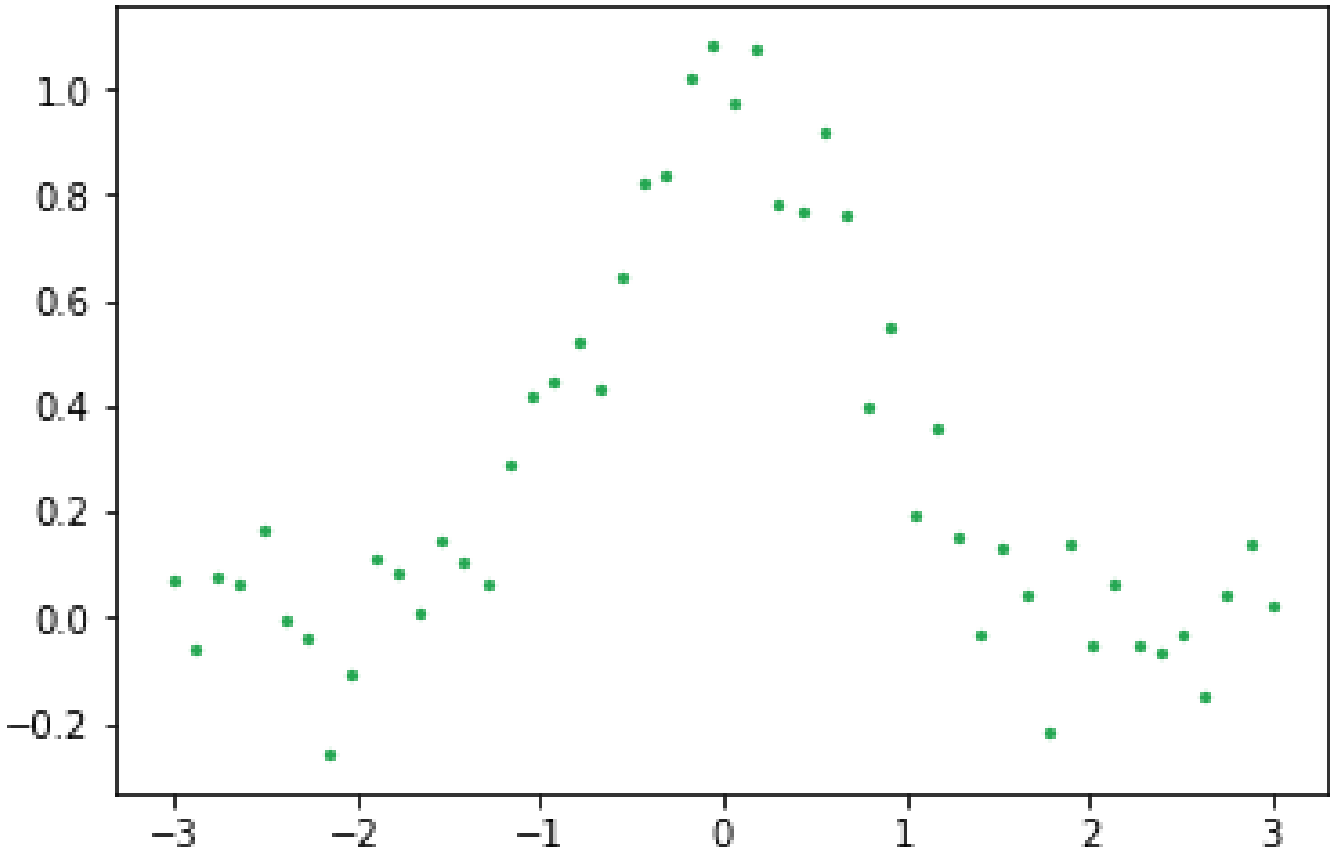
The Problem

Splines – piecewise polynomial functions - great for modeling non global data, but behave wildly near and outside their bounds

We want a spline model :

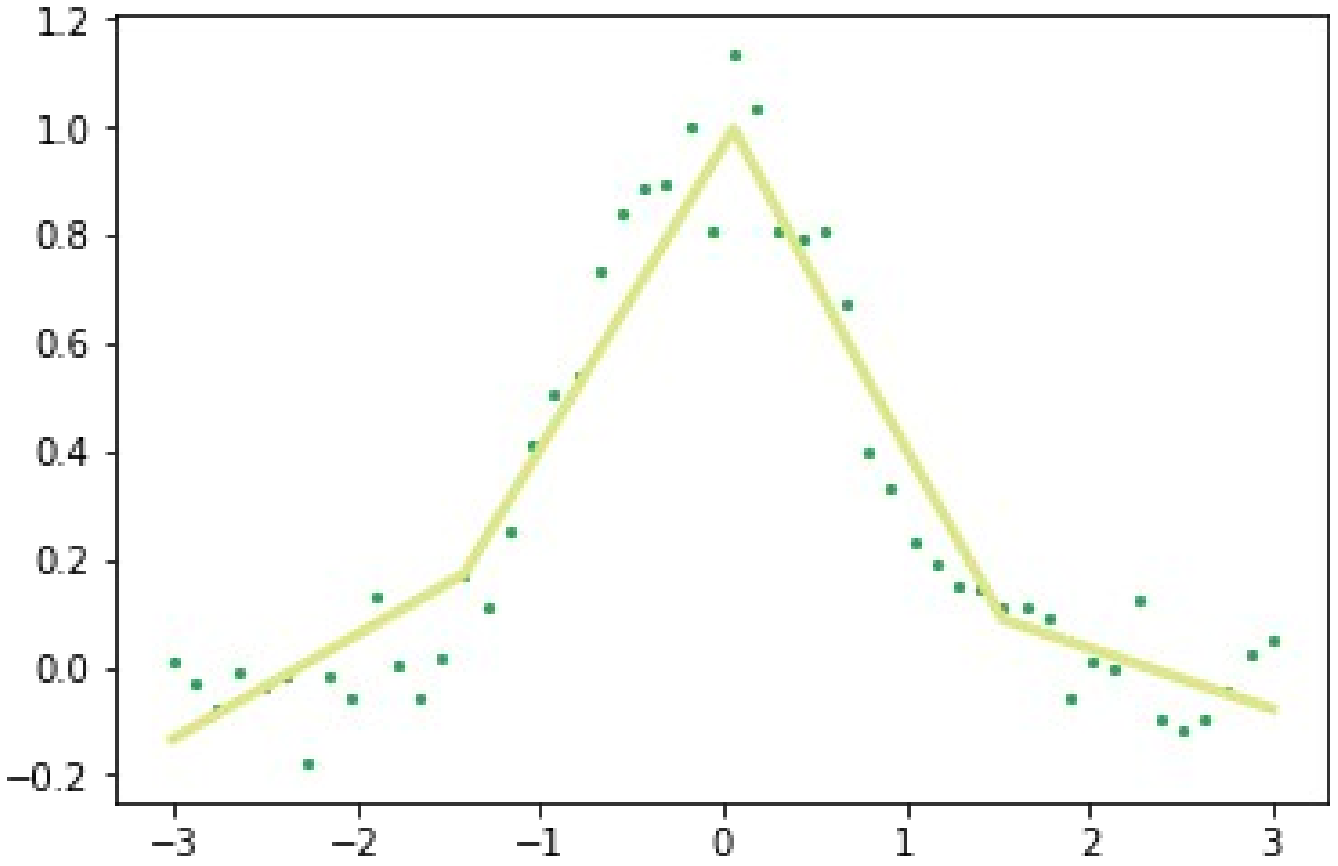
- to have nice behavior for better extrapolation
- to be fit without having to specify knot locations (optimal knots)
- to be as interpretable as possible (minimal knots/variable selection)
- and to have low error and be accurate

Splines



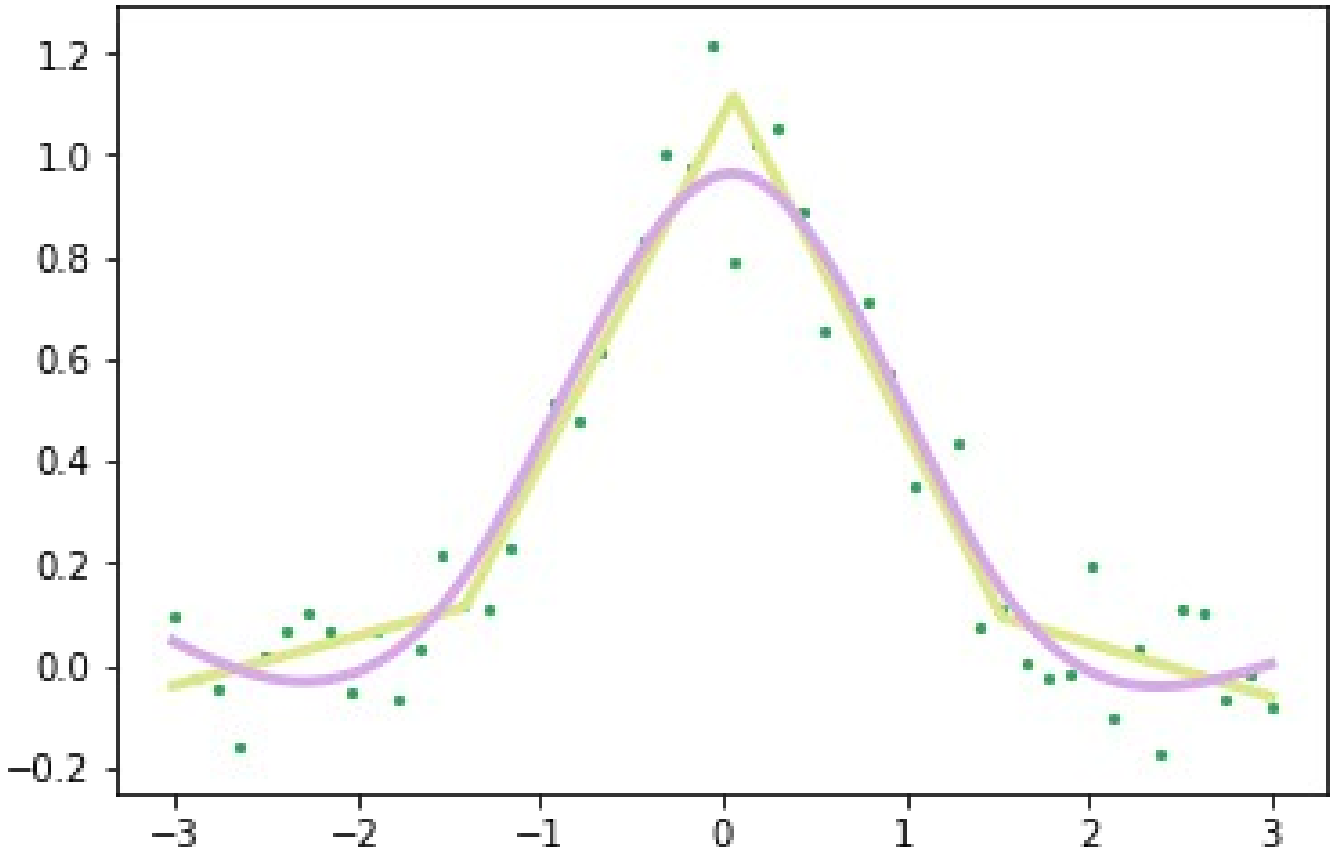
Splines

$$f(x) = c + \Sigma_{i=1}^{K} w_i (x - t_i)_+$$



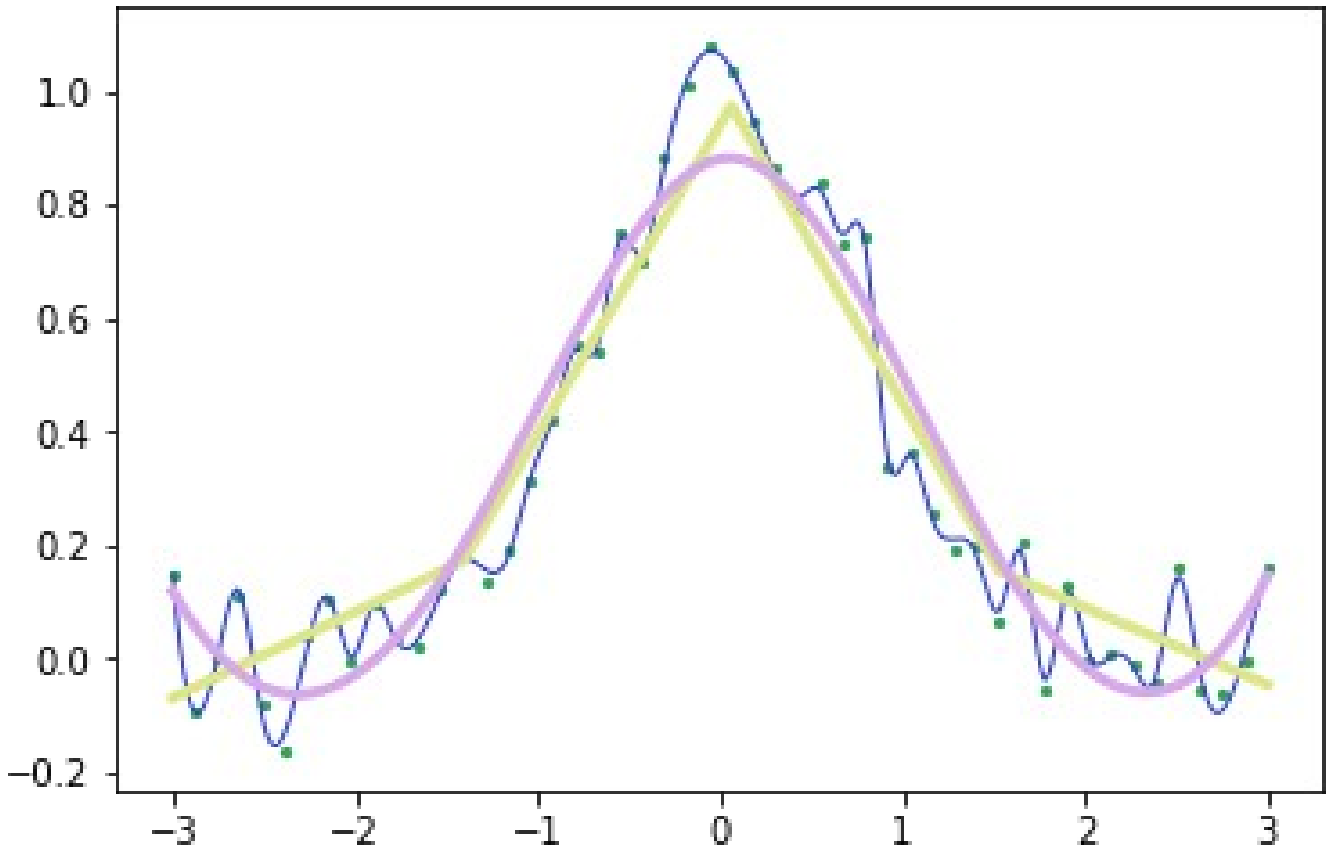
Splines

$$f(x) = c + \Sigma_{i=1}^{K} w_i (x - t_i)_+^3$$



Splines

$$f(x) = c + \Sigma_{i=1}^{K} w_i (x - t_i)_+^3$$



Background

Adaptive regressive splines" use l_1, sparse set of knots, but behaves crazy"
near/outside of bounds

Why does adding these kind of constraints help? It frees up degrees of freedom so you can add more knots, and it controls the behavior outside the bounds. (Some sort of belief about smoothness too)

What does it cost? Some bias near the boundaries, but this is where we know least anyways

	Smoothing Splines	Adaptive Regression Splines
Optimal knot selection	* * Gridding	
Behaves outside boundary		
Sparse knots		
References	De Boor 2001 Wahba 1990 Green 1994	Mammen and van de Geer 1997

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Behaves outside boundary			
Sparse knots			
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* Gridding

Atomic Measures

$$\mu = \Sigma_{j=1}^{K} w_j \delta_{t_j}$$

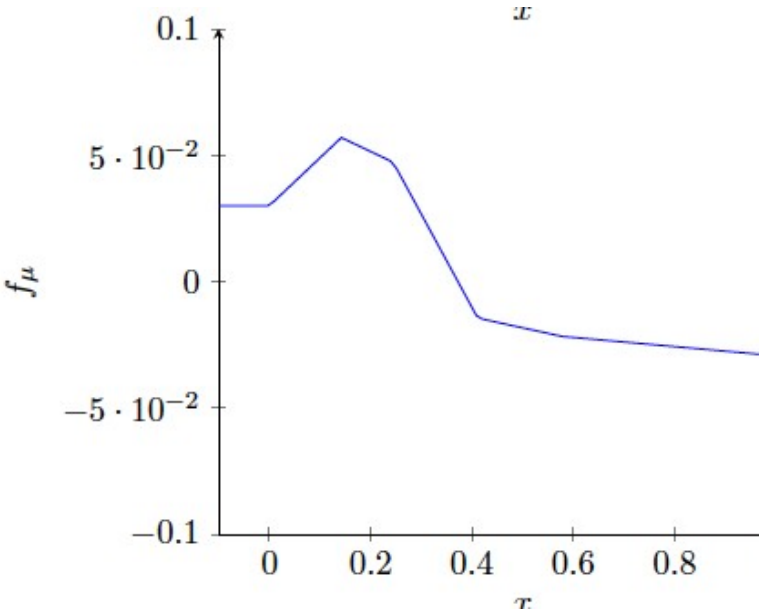
Saturation Constraint

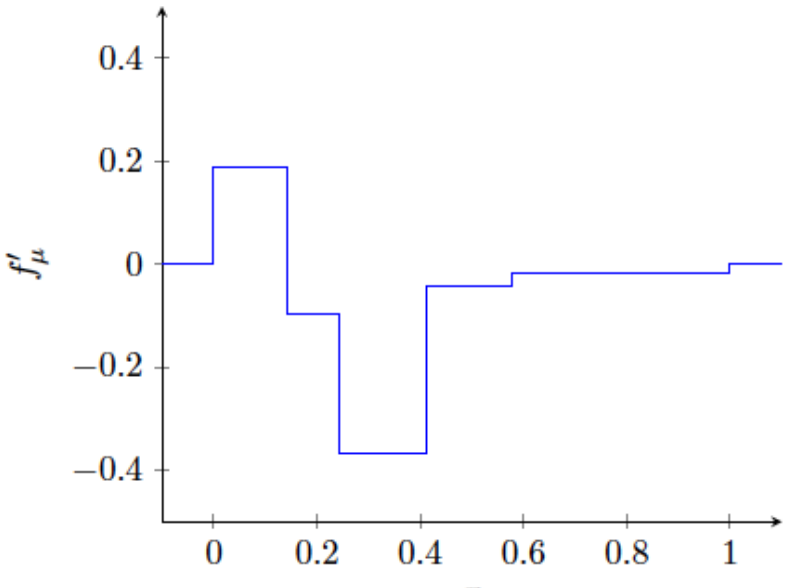
Saturating – constant on both intervals outside the range

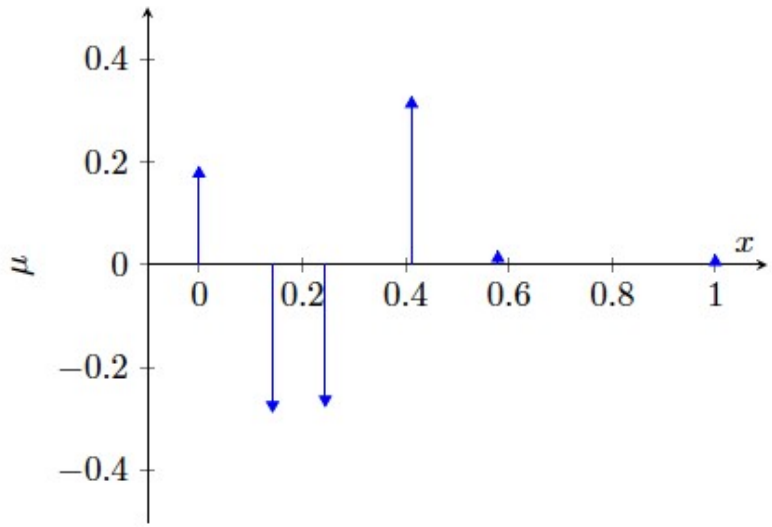
$$f'(x) = 0 \text{ for } x \notin [0, 1]$$

$$f(x) = f(0) \text{ for } x \leq 0$$

$$f(x) = f(1) \text{ for } x \geq 1$$


x
0.1
$5 \cdot 10^{-2}$
0
$-5 \cdot 10^{-2}$
-0.1
f_μ
0
0.2
0.4
0.6
0.8
x


0.4
0.2
0
-0.2
-0.4
f'_μ
0
0.2
0.4
0.6
0.8
1

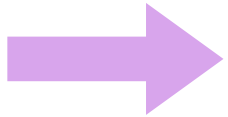

0.4
0.2
0
-0.2
-0.4
μ
x
0
0.2
0.4
0.6
0.8
1

General Optimization Problem

$$\min \Sigma_{i=1}^{n} l(f(x_i), y_i)$$

$$\int |f''(x)| dx \leq c$$

$$f'(x) = 0 \; x \notin [0,1]$$



$$\min L(E_x \mu + c)$$

$$||\mu||_1 \leq c$$

$$\int d\mu = 0$$

Optimization problem

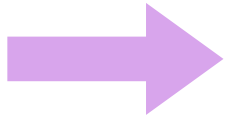
Solutions are measures

General Optimization Problem

$$\min L(E_x\mu + c)$$

$$||\mu||_1 \leq t$$

$$\int d\mu = 0$$



$$\min \int \langle g, \phi(t) \rangle ds(t)$$

$$||s||_1 \leq t$$

$$\int ds = 0$$

Conditional gradient descent

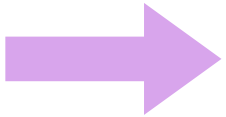
General Optimization Problem

$$\min L(E_x\mu + c)$$

$$||\mu||_1 \leq \tau$$

$$\int d\mu = 0$$

$$\mu_{supp} \subset \{t_1, \ldots, t_{2K}\}$$

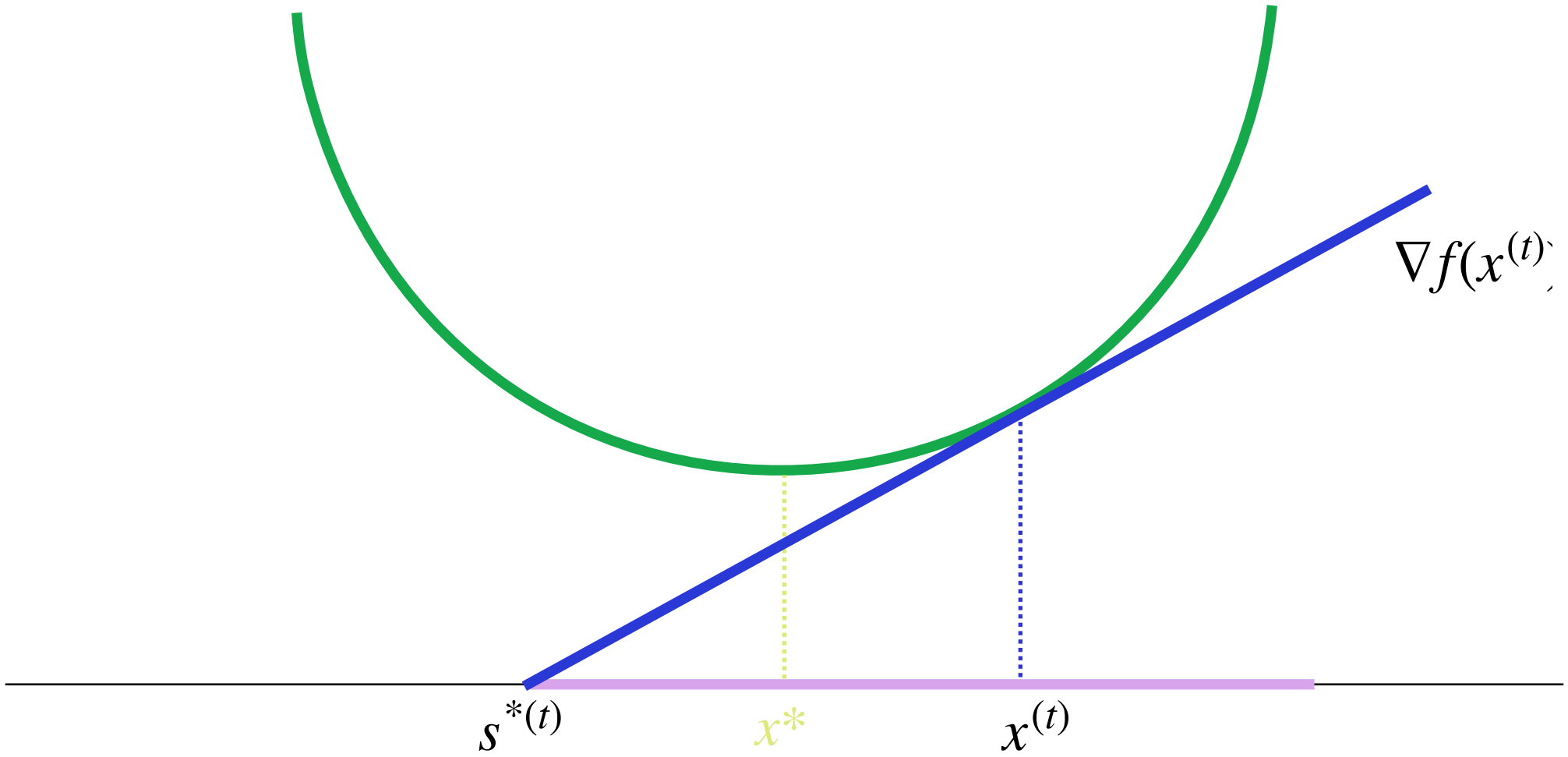


$$\min L(\Sigma_j w_j E_x \delta_{t_j})$$

$$||w||_1 \leq \tau$$

$$\mathbf{1}^T w = 0$$

Conditional Gradient Descent

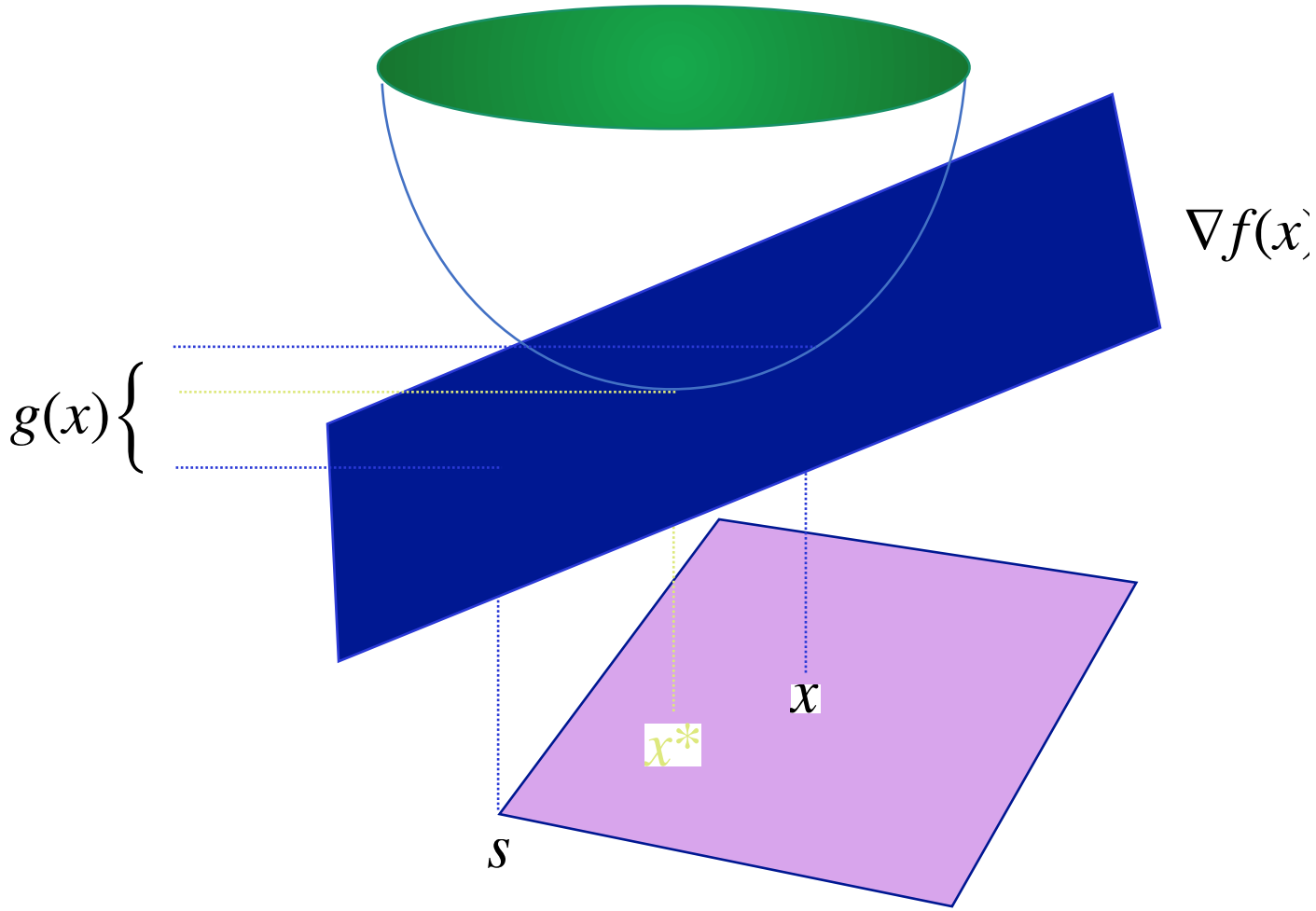


Conditional Gradient Descent

$$s := \arg\min_{s \in D} \langle s, \nabla f(x) \rangle$$

$$g(x) := \max_{s \in D} \langle x - s, \nabla f(x) \rangle$$

$$f(x) - f(x^*) \leq g(x)$$



$$f(x^{(t)}) - f(x^*) \leq O(\frac{1}{t})$$

Conditional Gradient Descent

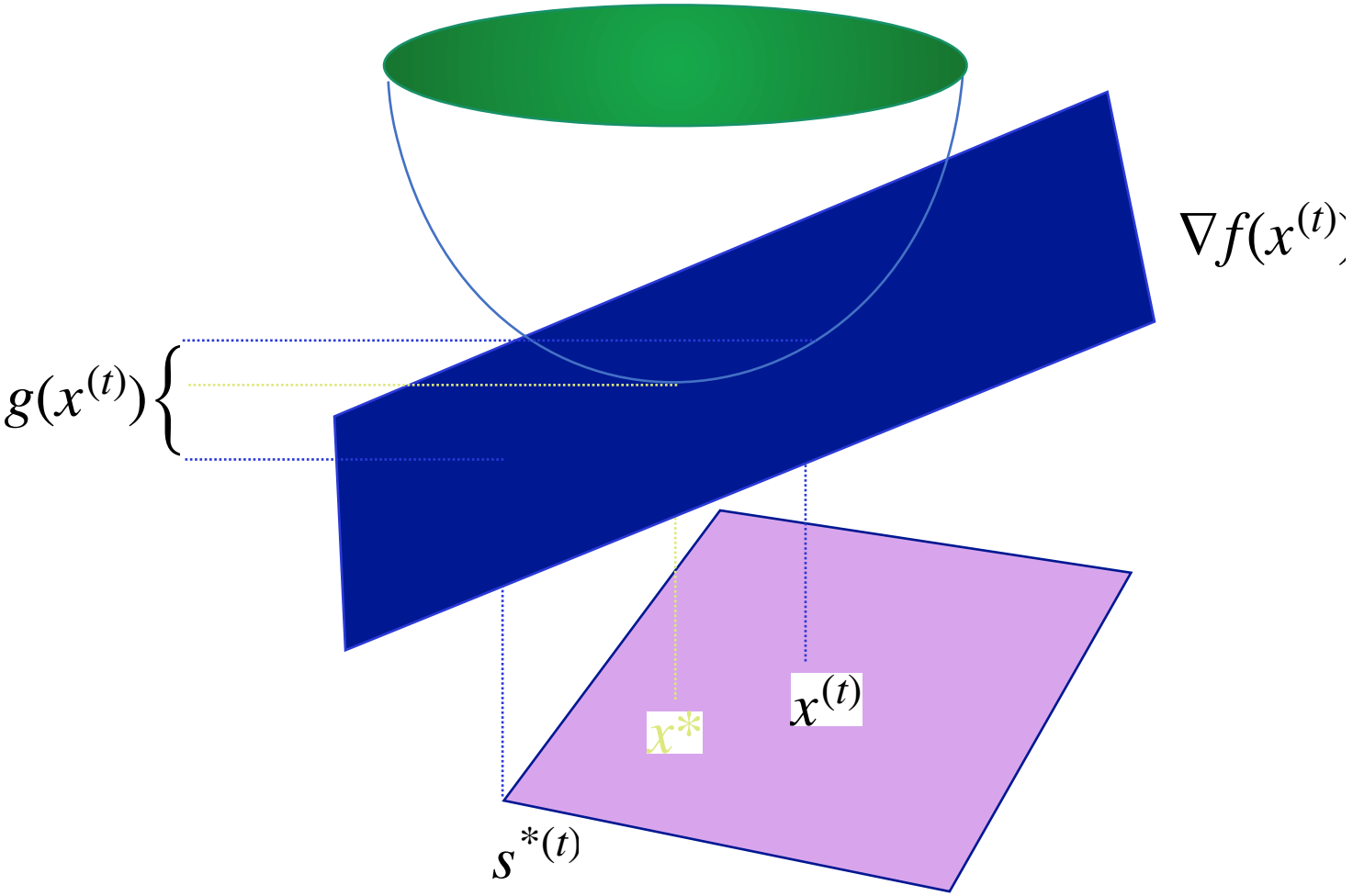
Algorithm 1 Frank-Wolfe

$$\langle s^{(t)}, \nabla f(x^{(t)}) \rangle \leq \langle s^{*(t)}, \nabla f(x^{(t)}) \rangle + \frac{\delta_\gamma(t) C_f}{2}$$

$$g(x^{(t)}) = \langle \nabla f(x^{(t)}), x^{(t)} - s^{(t)} \rangle$$

$$\gamma^{(t)} = \arg \min_{\gamma \in [0,1]} f(x^{(t)} + \gamma d_t)$$

$$x^{(t+1)} := \arg \min_{x \in conv(s^{(0)},..,s^{(t+1)})} f(s)$$



GAMs

$$g(\mu) = \alpha + \Sigma_{i=1}^{d} f_i(X_i)$$

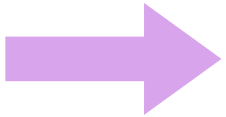
General Optimization Problem

$$\min L(E_x\mu + c)$$

$$||\mu||_1 \leq \tau$$

$$\int d\mu = 0$$

$$\mu_{supp} \subset \{t_1, \ldots, t_{2K}\}$$



$$\min L(\Sigma_j w_j E_x \delta_{t_j})$$

$$||w||_1 \leq \tau$$

$$\mathbf{1}^T w = 0$$

Experiments - Data

Adaptive regressive splines” use l 1, sparse set of knots, but behaves crazy“ near/outside of bounds

Why does adding these kind of constraints help? It frees up degrees of freedom so you can add more knots, and it controls the behavior outside the bounds. (Some sort of belief about smoothness too)

What does it cost? Some bias near the boundaries but this is where we know the least anyways

	Predicting	Covariates	Observations
ABALONE	Age	8 physical characteristic	4177
SPAM	Label (Spam/Not)	57 word frequency vectors	4601
BONE DENSITY			
ALS			

Experiments

ABALONE – Standard adaptive splines: Comparable RMSE, fewer knots

SPAM – Standard adaptive splines: Much better performance

Smoothing splines(GLMNet): Slightly better error, fewer knots

BONE DENSITY - Smoothing splines: Fewer knots, more robust to outliers (algorithm can fit general loss functions)

ALS – Boosted decision trees, GLMNet, and random forests: Comparable performance, far fewer features, no interaction terms

Curvature Constant

$$C_f := \sup_{x,s \in D, \gamma \in [0,1], x' = x + \gamma(s-x)} \frac{2}{\gamma^2}(f(x') - f(x) - \langle x' - x, \nabla f(x) \rangle)$$

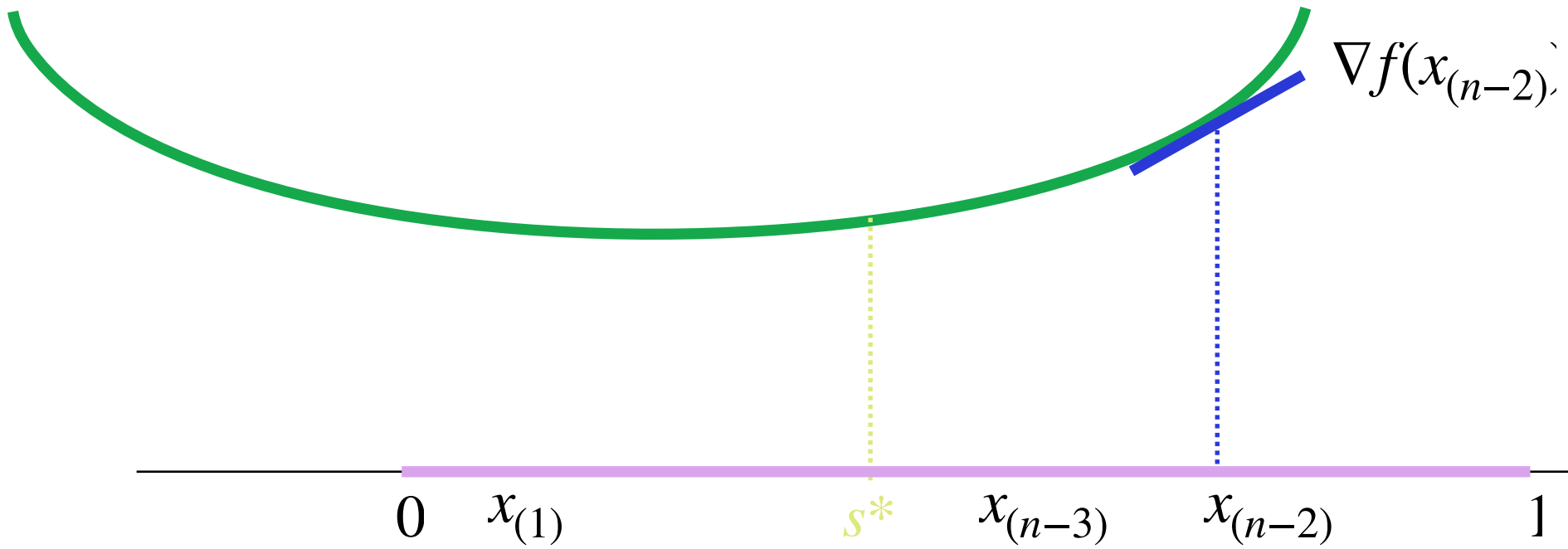
L-Lipschitz continuous gradient

$$||\nabla f(x) - \nabla f(y)|| \leq L||x - y|| \forall x, y \qquad L > 0$$

Does not require convexity!

Questions

Conditional Gradient Descent



$$s_{(n-2)} = \nabla f(x_{(n)})(x_{(n)} - x_{(n-1)})$$

$$s_{(n-3)} = \nabla f(x_{(n)})(x_{(n)} - x_{(n-2)}) + \nabla f(x_{(n-1)})(x_{(n-1)} - x_{(n-2)})$$

$$s_{(n-3)} = s_{(n-2)} + (\nabla f(x_{(n)} + \nabla f(x_{(n-1)})(x_{(n-1)} - x_{(n-2)})$$

Higher-order splines

Smoothing Splines

$$RSS(f,\lambda) = \Sigma_{i=1}^{N}\{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 dt$$

$$f(x) = \Sigma_{j=1}^{N} N_j(x)\theta_j$$

$$RSS(\theta,\lambda) = (y - N\theta)^T(y - N\theta) + \lambda\theta^T\Omega_N\theta$$

$$\hat{\theta} = (N^T N + \lambda\Omega_N)^{-1} N^T y$$

$$\hat{f}(s) = \Sigma_{j=1}^{N} N_j(x)\hat{\theta}_j$$

What new ideas do the authors .3 ?propose

Developed a conditional gradient-based algorithm to fit saturating splines without pre-specified knot locations (for splines of degree at least 2), an infinite-dimensional problem

When applied to additive models, saturation splines result in variable selection and function fitting

Original Problem

$$\text{minimize } \Sigma_{i=1}^{n} l(f(x_i), y_i)$$

$$\text{subject to } \int |f''(x)| dx \leq \tau$$

New Problem

$$\text{minimize } \Sigma_{i=1}^{n} l(E_x \mu + c, y_i)$$

$$\text{subject to } ||\mu||_1 \leq \tau$$

$$\int d\mu = 0$$

Newer Problem

$$\text{minimize} \int \langle \nabla L(E_x \mu), \psi(t) \rangle ds(t)$$

$$\text{subject to } ||s||_1 \leq \tau$$

$$\int ds = 0$$